Python

--- Anaconda

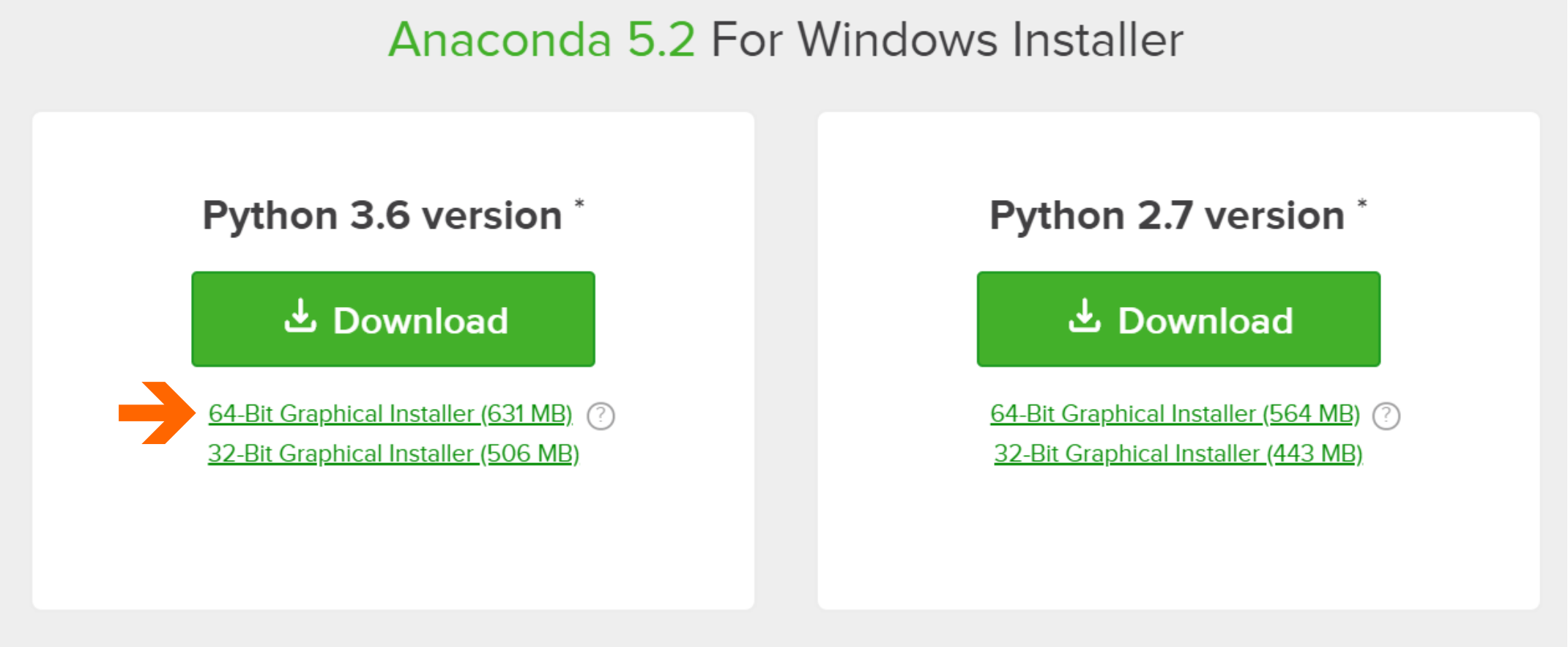
--- Jupyter, etc.

高怡宣
陽明大學 生物醫學影像暨放射科學系

python

ANACONDA®

下載 Anaconda ➔ https://www.anaconda.com/download/



Install Anaconda, Jupyter Notebook

安裝 Anaconda, Python, Jupyter Notebook, Tensorflow, Keras

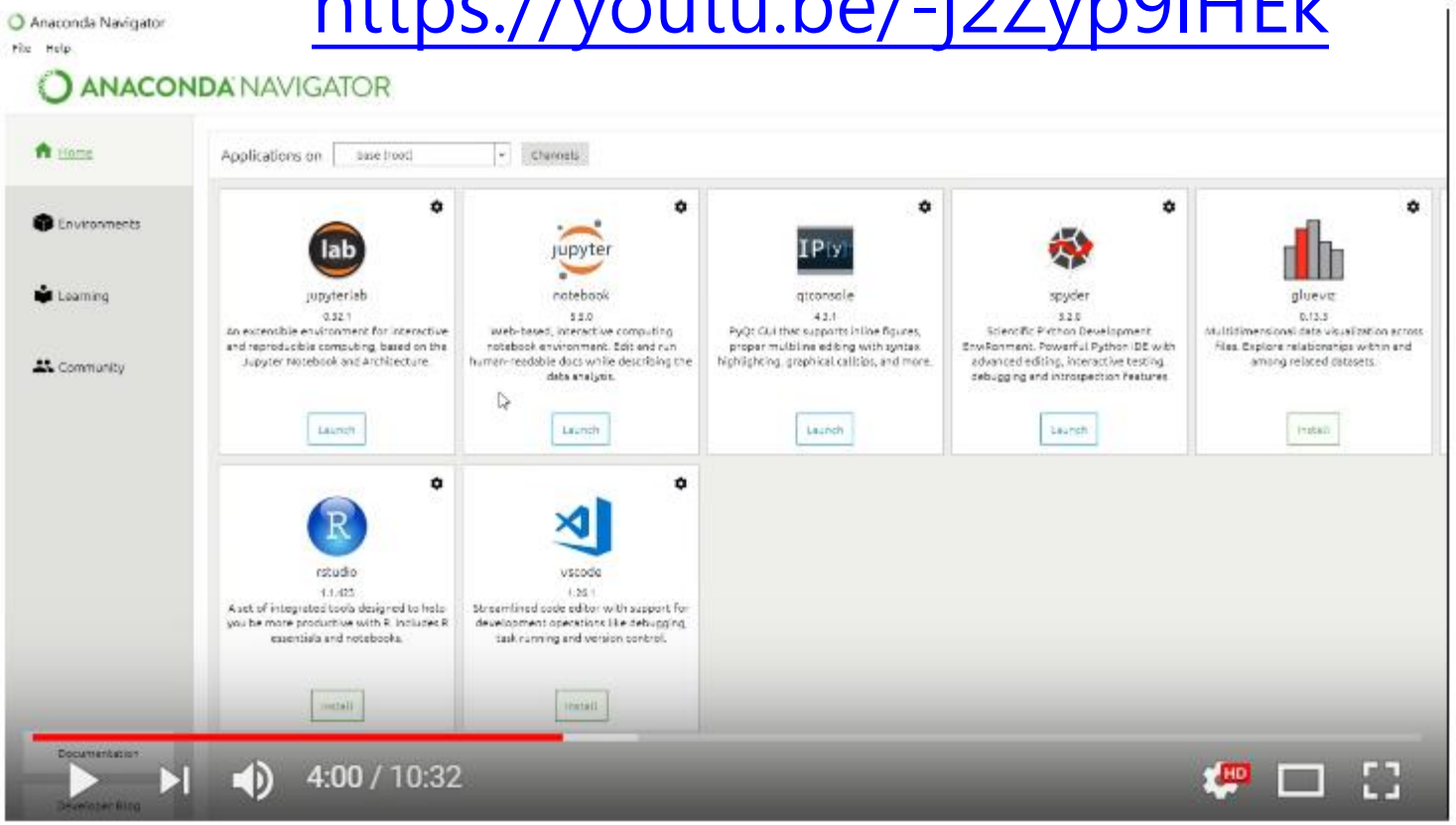
https://youtu.be/-j2Zyp9lHEk

Jupyter Notebook 使用及安裝介紹

https://www.youtube.com/watch?v=FMGyLc2sm88

YouTube 搜尋

https://youtu.be/-j2Zyp9lHEk



安裝 Anaconda, Python, Jupyter Notebook, Tensorflow, Keras

觀看次數：0

0 0 分享

高怡宣老師
發佈日期：2018年8月27日

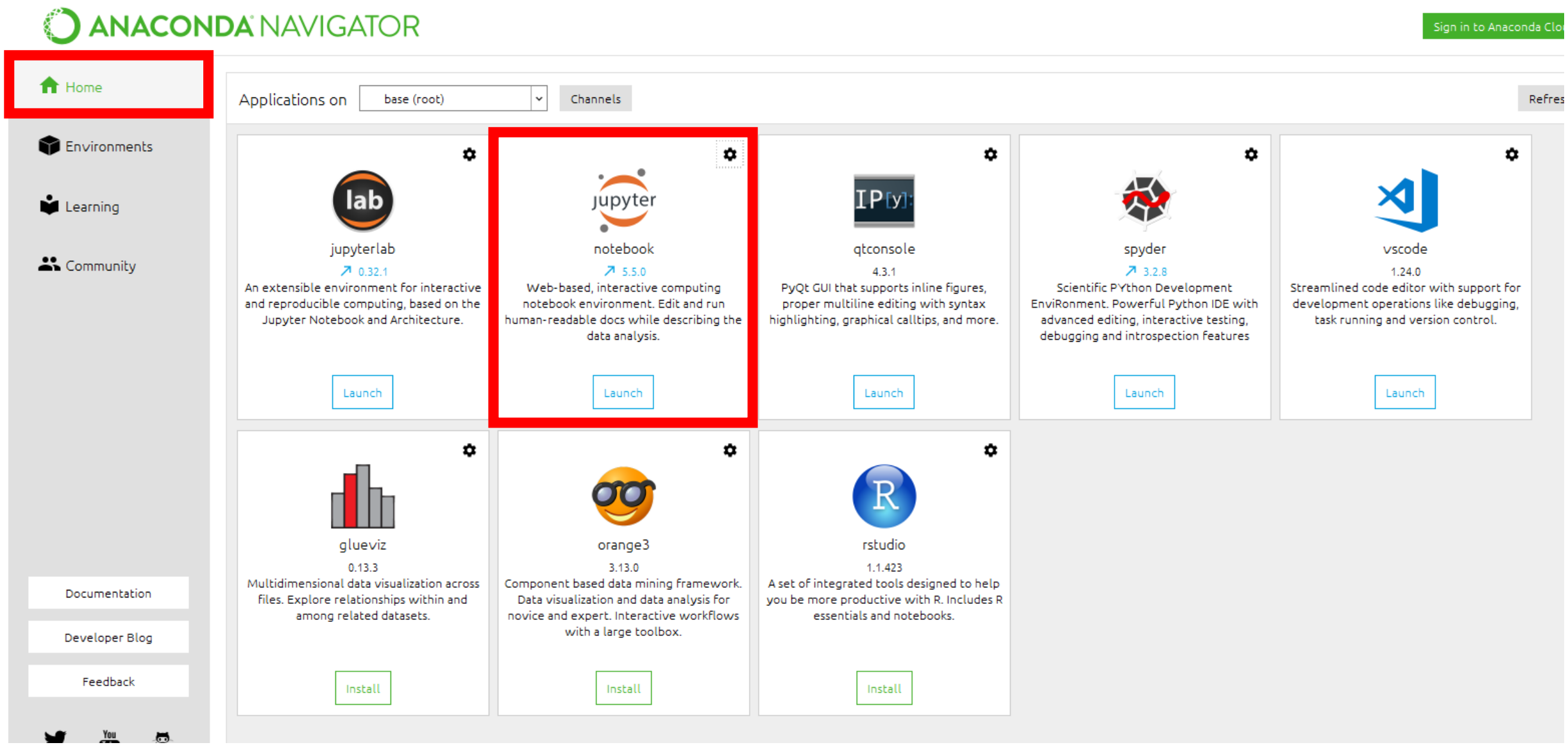
數據分析 編輯影片

In Anaconda prompt:

1. conda update conda
2. conda update jupyter

In Anaconda ➔ environment

1. Numpy
2. Matplotlib
3. Seaborn
4. Pandas
5. Tensorflow
6. Keras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Channels
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jupyterlab
0.32.1
An extensible environment for interactive and reproducible computing, based on the Jupyter Notebook and Architecture.
Launch
notebook
5.5.0
Web-based, interactive computing notebook environment. Edit and run human-readable docs while describing the data analysis.
Launch
qtconsole
4.3.1
PyQt GUI that supports inline figures, proper multiline editing with syntax highlighting, graphical calltips, and more.
Launch
spyder
3.2.8
Scientific PYthon Development EnviRonment. Powerful Python IDE with advanced editing, interactive testing, debugging and introspection features
Launch
vscode
1.24.0
Streamlined code editor with support for development operations like debugging, task running and version control.
Launch
glueviz
0.13.3
Multidimensional data visualization across files. Explore relationships within and among related datasets.
Install
orange3
3.13.0
Component based data mining framework. Data visualization and data analysis for novice and expert. Interactive workflows with a large toolbox.
Install
rstudio
1.1.423
A set of integrated tools designed to help you be more productive with R. Includes R essentials and notebooks.
Install

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base (root)

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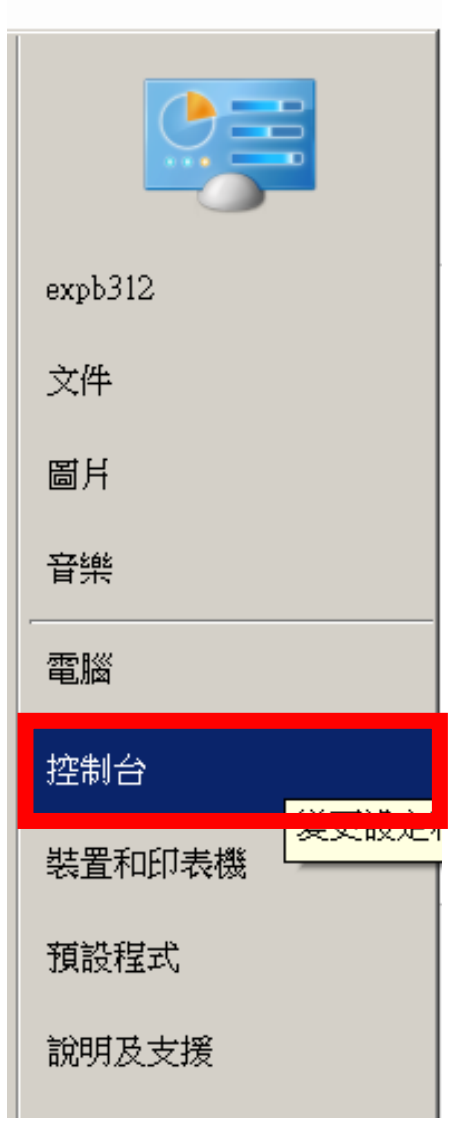
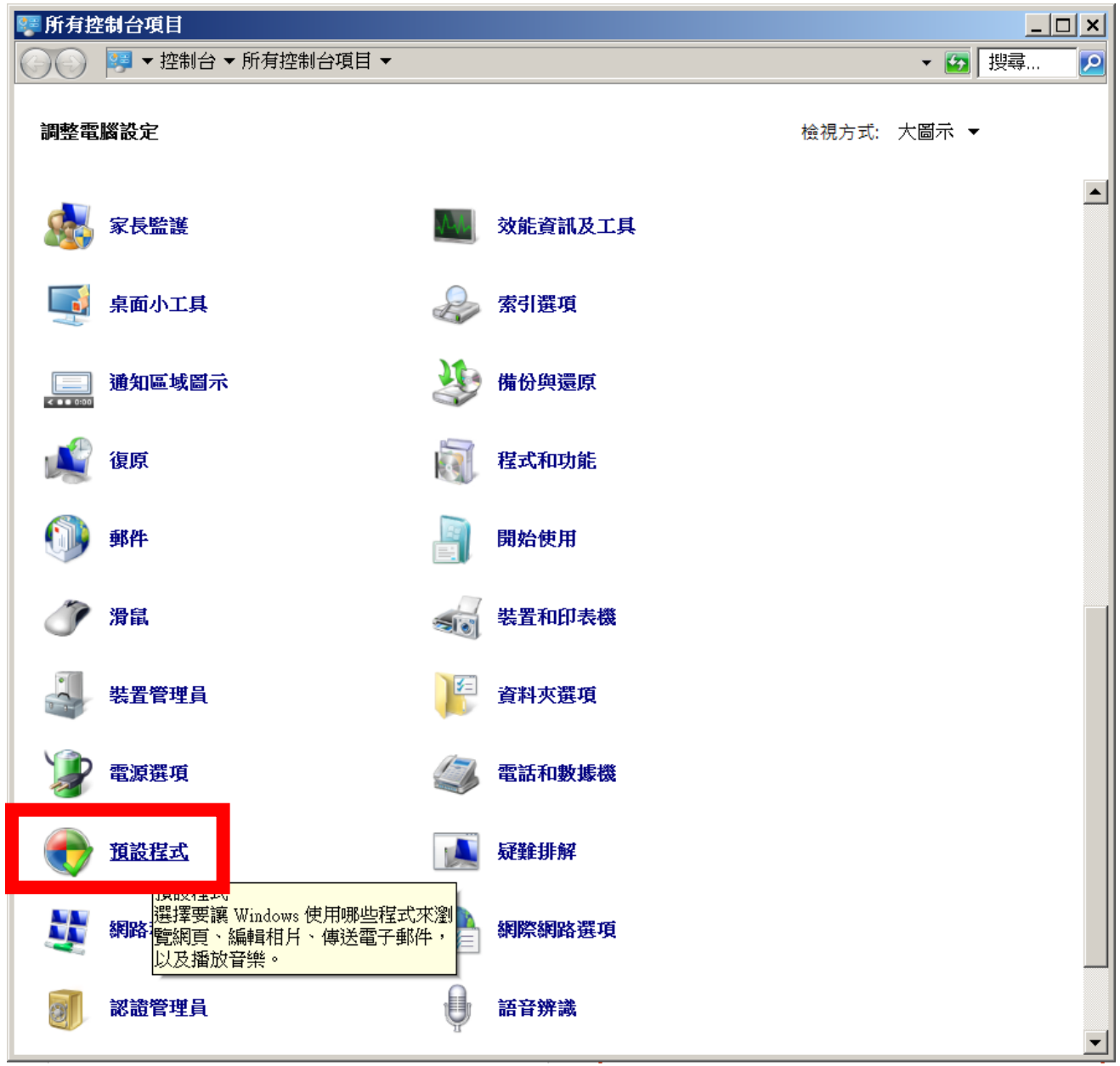
Name	T	Description
nltk		Build python programs to work with human language data
nose		Nose extends unittest to make testing easier
notebook		Web-based notebook environment for interactive computing
numba		Numpy aware dynamic python compiler using llvm
numexpr		Fast numerical expression evaluator for numpy
numpy		Array processing for numbers, strings, records, and objects
numpy-base		
numpydoc		Sphinx extension to support docstrings in numpy format
odo		Shapeshifting for your data
olefile		Parse, read and write microsoft ole2 files
openpyxl		A python library to read/write excel 2010 xlsx/xlsm files
openssl		Openssl is an open-source implementation of the ssl and tls protocols


expb312
文件
圖片
音樂
電腦
控制台
裝置和印表機
預設程式
說明及支援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控制台 ▾ 所有控制台項目 ▾
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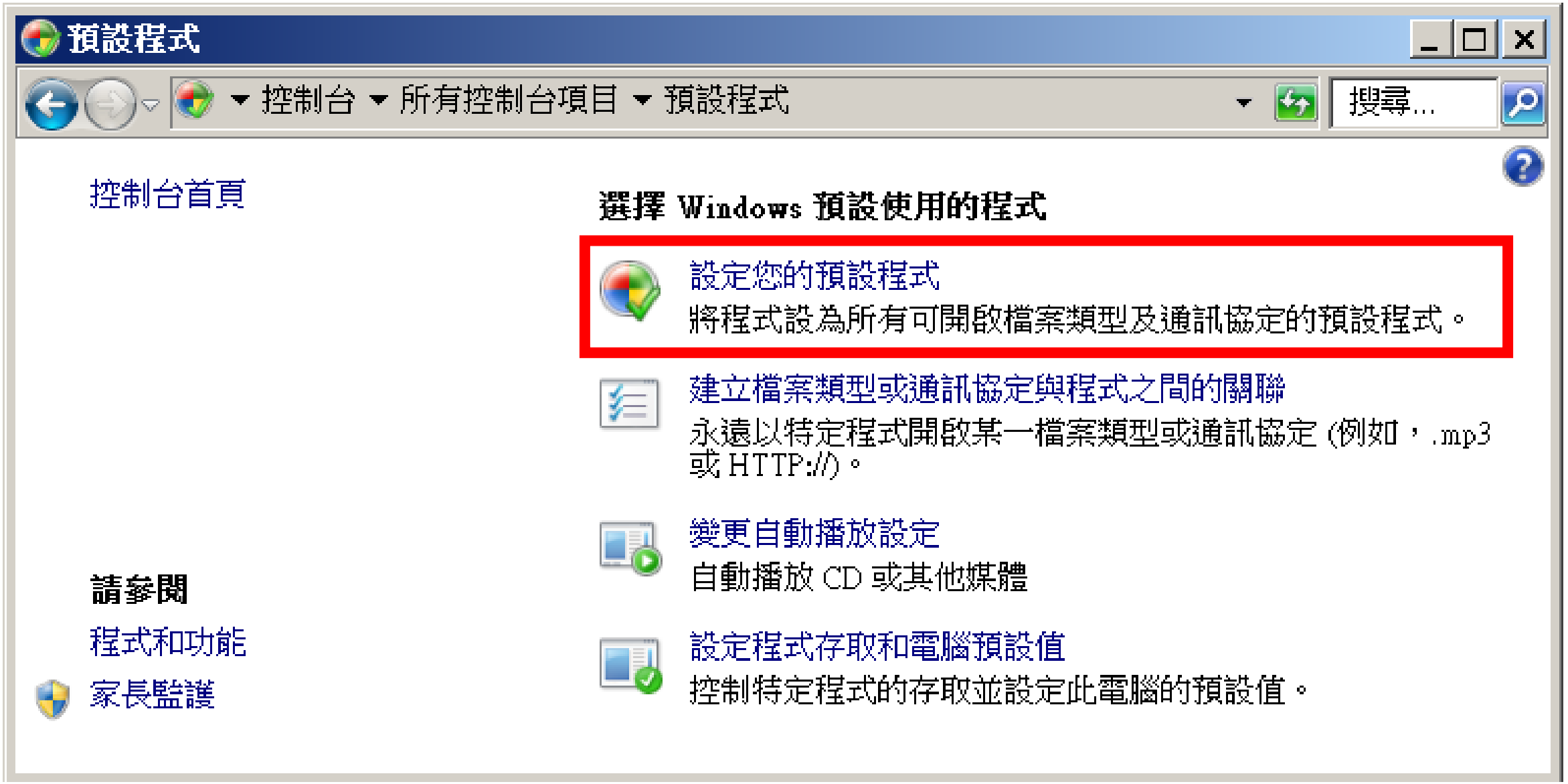
調整電腦設定
檢視方式: 大圖示 ▾
家長監護
效能資訊及工具
桌面小工具
索引選項
通知區域圖示
備份與還原
復原
程式和功能
郵件
開始使用
滑鼠
裝置和印表機
裝置管理員
資料夾選項
電源選項
電話和數據機
預設程式
疑難排解
選擇要讓 Windows 使用哪些程式來瀏覽網頁、編輯相片、傳送電子郵件，以及播放音樂。
網際網路選項
認證管理員
語音辨識


預設程式
控制台 ▾ 所有控制台項目 ▾ 預設程式
搜尋...
控制台首頁
選擇 Windows 預設使用的程式
設定您的預設程式
將程式設為所有可開啟檔案類型及通訊協定的預設程式。
建立檔案類型或通訊協定與程式之間的關聯
永遠以特定程式開啟某一檔案類型或通訊協定 (例如，.mp3 或 HTTP://)。
變更自動播放設定
自動播放 CD 或其他媒體
設定程式存取和電腦預設值
控制特定程式的存取並設定此電腦的預設值。
請參閱
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家長監護

設定您的預設程式

若要將程式設為所有可開啟檔案類型及通訊協定的預設程式，請按一下程式，然後按一下 [設定成預設值]。

程式集

- Access (桌面)
- Adobe Acrobat Reader DC
- Excel (桌面)
- Google Chrome
- InfoPath (桌面)
- Internet Explorer
- Lync (桌面)
- MATLAB (R2013b)
- OneNote (桌面)
- Outlook (桌面)
- PowerPoint (桌面)
- Publisher (桌面)
- Windows Media Center
- Windows Media Player
- Windows Search 總管
- Windows 光碟映像燒錄程式
- Windows 相片檢視器
- Windows 連絡人

Google Chrome
Google Inc

Google Chrome 瀏覽器開啟網頁和執行應用程式的速度奇快無比！除了執行速度迅速、穩定且容易使用之外，內建防護機制還可讓你安心瀏覽網頁，無需擔心受到網路釣魚與惡意軟體的威脅。

此程式全部使用預設值

將此程式設定成預設值(S)
使用選取的程式來開啟預設可開啟的所有檔案類型和協定。

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選擇檔案類型和協定，以便預設由選取的程式開啟。

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搜尋...

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選取預設由此程式開啟的副檔名，然後按一下 [儲存]。

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☑ .html	Chrome HTML Document	Google Chrome
☐ .pdf	Adobe Acrobat Document	Adobe Acrobat Reader DC
☑ .shtml	Chrome HTML Document	Google Chrome
☑ .svg	Chrome HTML Document	Google Chrome
☑ .webp	Chrome HTML Document	Google Chrome
☑ .xht	Chrome HTML Document	Google Chrome
☑ .xhtml	Chrome HTML Document	Google Chrome
通訊協定		
☑ FTP	Chrome HTML Document	Google Chrome
☑ HTTP	Chrome HTML Document	Google Chrome
☑ HTTPS	Chrome HTML Document	Google Chrome

儲存　取消

The Jupyter Notebook

The Jupyter Notebook is an open-source web application that allows you to **create and share documents that contain live code, equations, visualizations and narrative text.** Uses include: data cleaning and transformation, numerical simulation, statistical modeling, data visualization, machine learning, and much more.

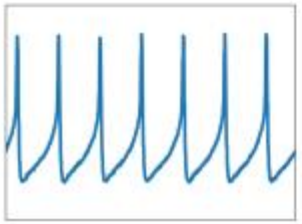
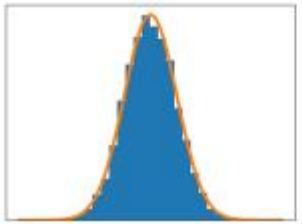
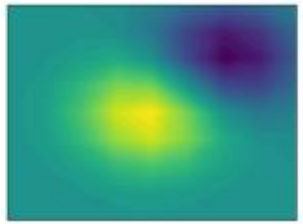
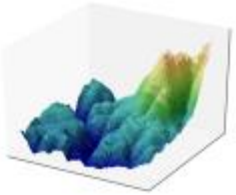
http://jupyter.org/

NumPy

NumPy is the fundamental package for scientific computing with Python. It contains: a powerful **N-dimensional array object, sophisticated functions, tools for integrating C/C++ and Fortran code, useful linear algebra, Fourier transform, and random number capabilities.**

http://www.numpy.org/

matplotlib

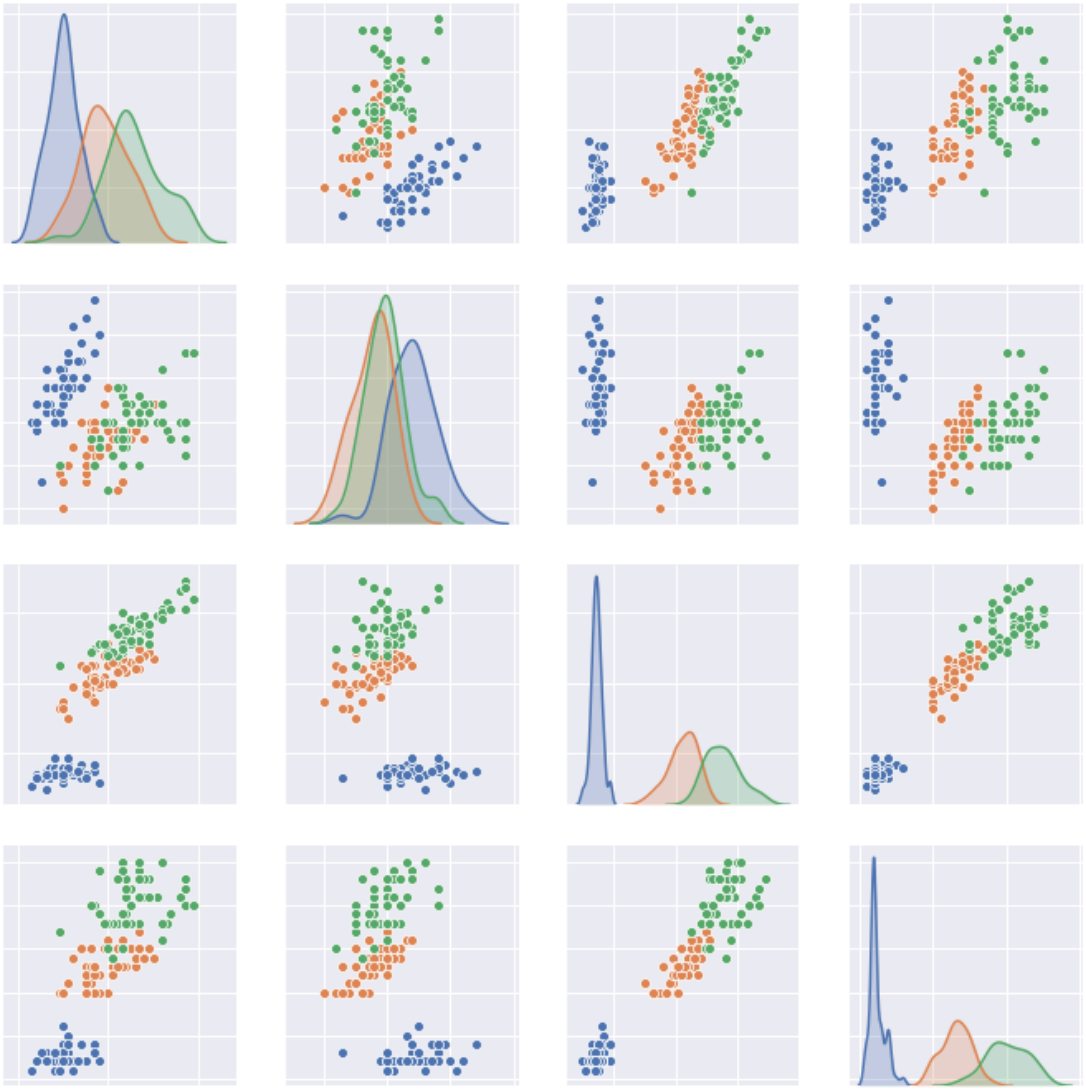





Matplotlib is a Python 2D plotting library which produces **publication quality figures** in a variety of hardcopy formats and interactive environments across platforms. Matplotlib can be used in Python scripts, the Python, the Jupyter notebook.

https://matplotlib.org/index.html

Seaborn

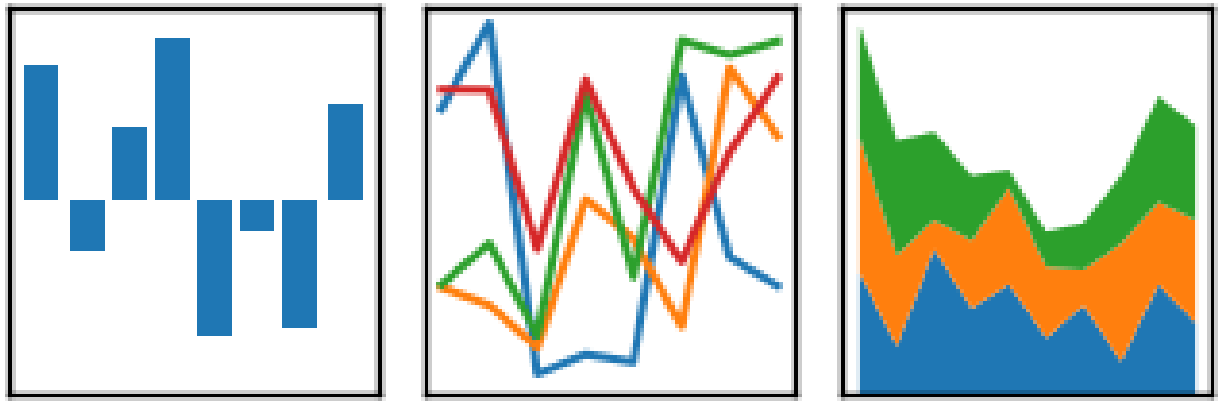
Seaborn is a library for making **statistical graphics** in Python. It is built on top of matplotlib and closely **integrated with pandas data structures.**



https://seaborn.pydata.org/

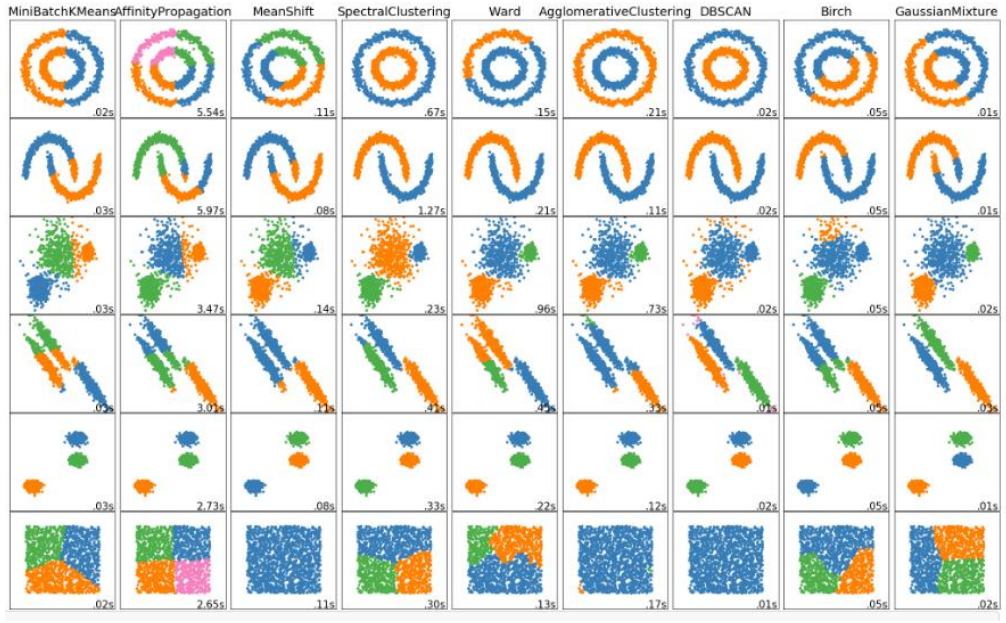
pandas

$$y_{it} = \beta' x_{it} + \mu_i + \epsilon_{it}$$



pandas is an open source, BSD-licensed library providing high-performance, **easy-to-use data structures and data analysis tools** for the Python programming language.

https://pandas.pydata.org/



scikit-learn - Machine Learning in Python

Simple and efficient tools for **data mining and data analysis.**
Built on NumPy, SciPy, and matplotlib. Functions include:
classification, regression, clustering, dimension reduction, model selection, preprocessing.

http://scikit-learn.org/

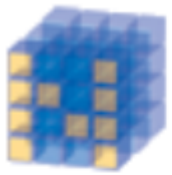
Keras

Keras is a high-level neural networks API, written in Python and capable of running on top of **TensorFlow, CNTK, or Theano**. It was developed with a focus on enabling fast experimentation. Being able to go from idea to result with the least possible delay is key to doing good research.

https://keras.io/

SciPy.org Sponsored By ENTHOUGHT

SciPy (pronounced “Sigh Pie”) is a Python-based ecosystem of open-source software for mathematics, science, and engineering. In particular, these are some of the core packages:



NumPy

Base N-dimensional array package

SciPy library

Fundamental library for scientific computing

Matplotlib

Comprehensive 2D Plotting

IP[y]: IPython

IPython

Enhanced Interactive Console

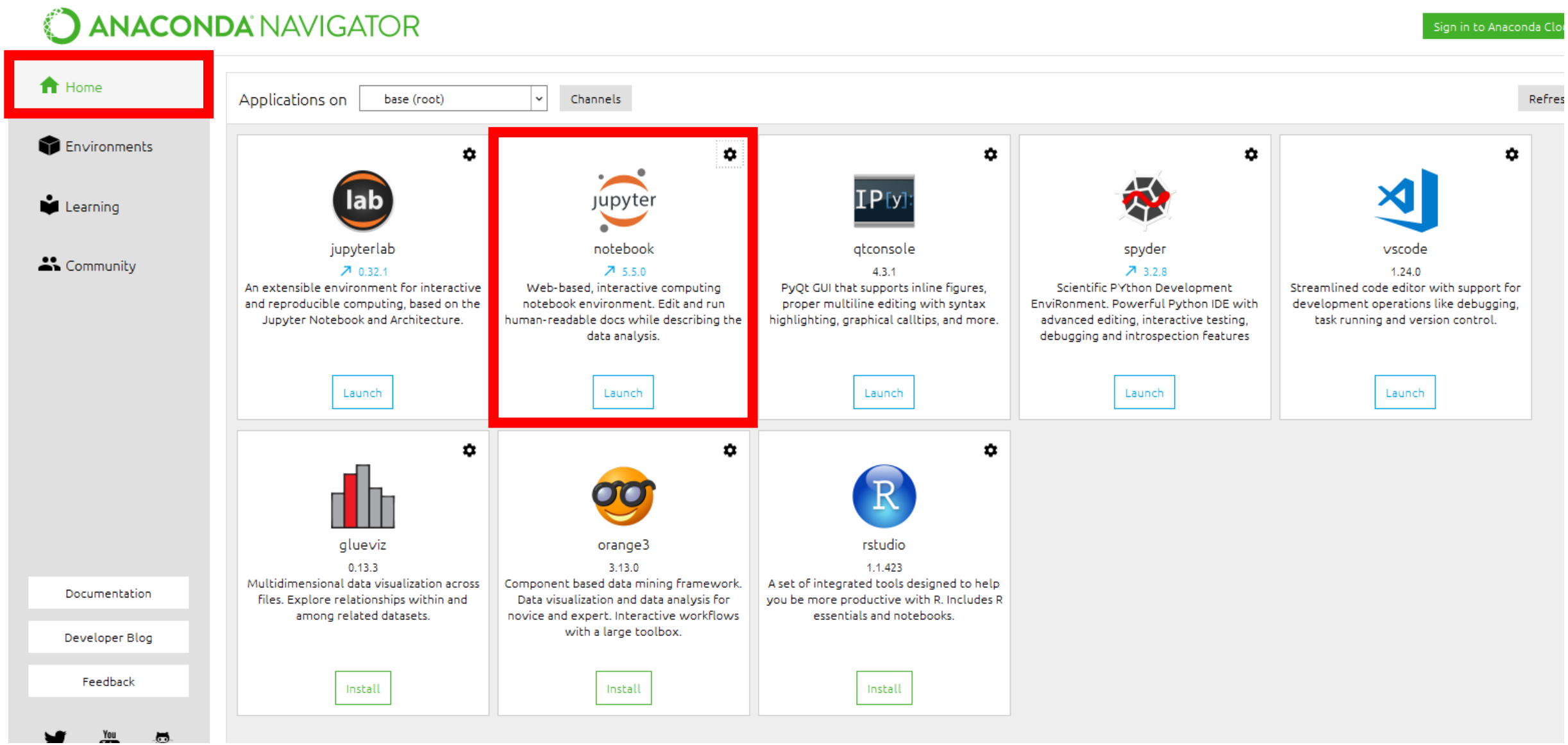
Sympy

Symbolic mathematics

pandas

Data structures & analysis

https://www.scipy.org/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spyder
3.2.8
Scientific PYthon Development EnviRonment. Powerful Python IDE with advanced editing, interactive testing, debugging and introspection features
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vscode
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glueviz
0.13.3
Multidimensional data visualization across files. Explore relationships within and among related datasets.
Install
orange3
3.13.0
Component based data mining framework. Data visualization and data analysis for novice and expert. Interactive workflows with a large toolbox.
Install
rstudio
1.1.423
A set of integrated tools designed to help you be more productive with R. Includes R essentials and notebooks.
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Files Running Clusters

Select items to perform actions on them.

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Python 3

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Create a new notebook with Pytl

Text File

Folder

Terminal

0 /

Name

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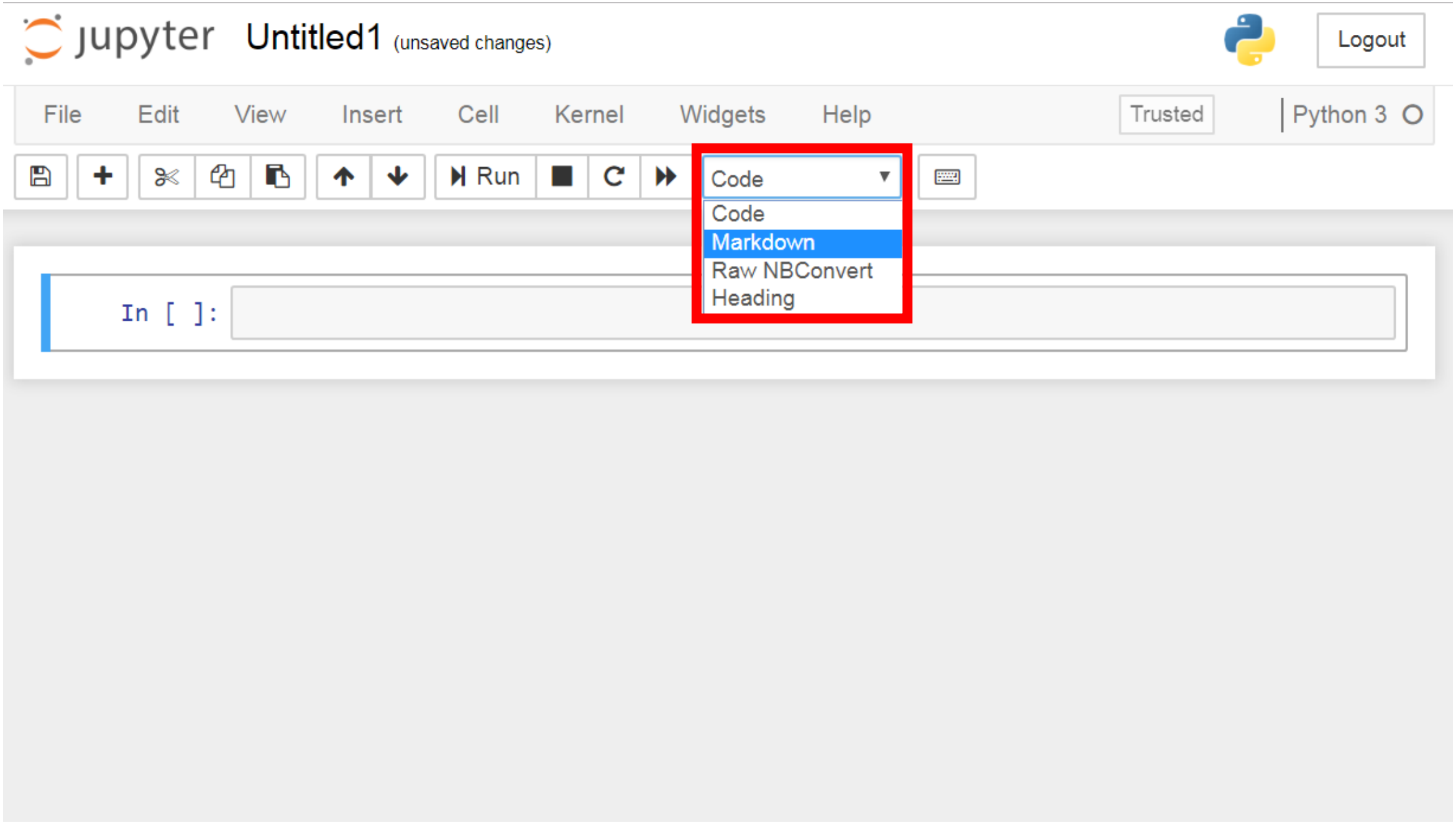
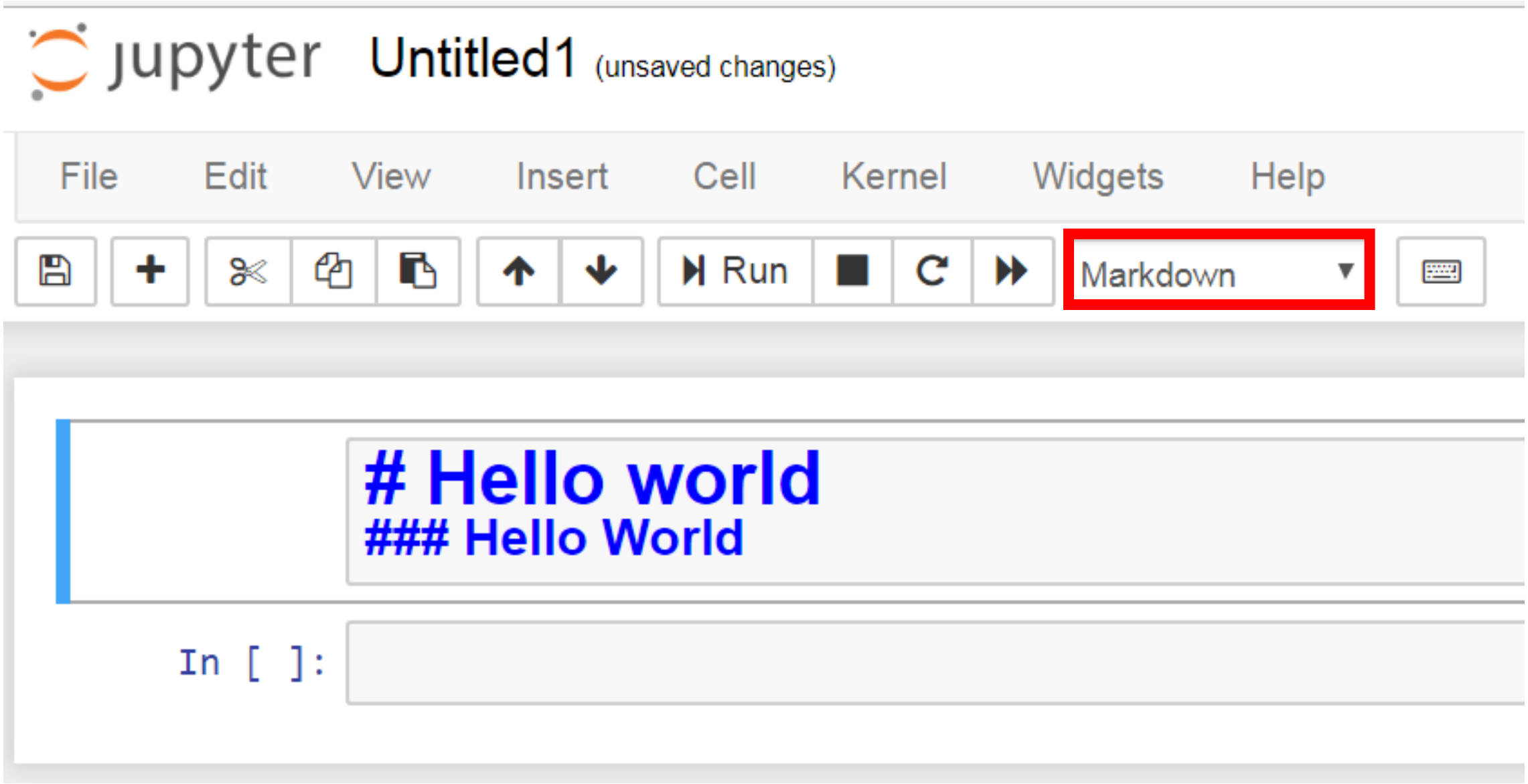
Google 雲端硬碟	3 年前
Links	9 個月前
Music	1 年前
Pictures	1 年前


jupyter
Untitled1 (unsaved changes)
Logout
File
Edit
View
Insert
Cell
Kernel
Widgets
Help
Trusted
Python 3
Run
Code
Code
Markdown
Raw NBConvert
Heading
In [ ]:


jupyter
Untitled1 (unsaved changes)
File
Edit
View
Insert
Cell
Kernel
Widgets
Help
Run
Markdown
Hello world
Hello World
In [ ]:

jupyter Untitled1 (unsaved changes)

File Edit View Insert Cell Kernel Widgets Help

Run Code

Hello world

Hello World

In [1]:
```
a=1
b=a+2
print(b)
```

```
3
```

Type the followings in Jupyter Notebook

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sn
import pandas as pd
import keras
import sklearn
```

In [11]:
```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sn
import pandas as pd
import keras
import sklearn
```

In [12]:
```
print("Good morning")
```

```
Good morning
```

In [13]:
```
a = 1
b = a + 3
print(b)
```

```
4
```

```
In [1]: import matplotlib.pyplot as plt
```

```
In [2]: x = [1,2,3,4,5]
        y = [5,1,4,2,3]
```

```
In [3]: plt.plot(x,y)
```

```
In [4]: for i in range(1,10):
            print(i)
```

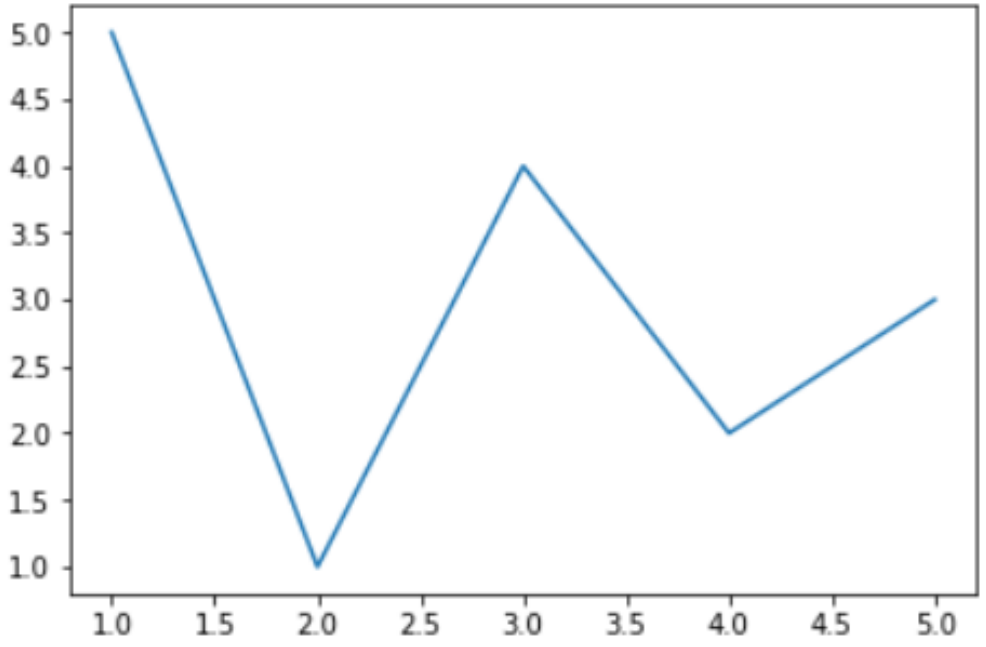
Jupyter Notebook使用及安装介绍 (0:00 ~ 4:20)
https://www.youtube.com/watch?v=FMGyLc2sm8

```
In [1]: import matplotlib.pyplot as plt

In [2]: x = [1,2,3,4,5]
        y = [5,1,4,2,3]

In [3]: plt.plot(x,y)

Out[3]: [<matplotlib.lines.Line2D at 0x81d1b70>]
```




```
In [4]: for i in range(1,10):
            print(i)

1
2
3
4
5
6
7
8
9
```

Jupyter Notebook使用及安装介绍 (0:00 ~ 4:20)
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Thank you for your attention